Design of a 4-bit Digital to Analog Converter

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Abstract

A Digital-to-Analog Converter is a circuit that turns digital into analog signals. The DAC type we are designing is called a weighted resistor D/A Converter Circuit. We designed this DAC with cost and simplicity in mind. This report has five sections, Introduction, 4-BIT Dac Design and Simulation, PCB Design, PCB Assembly and Testing, and 4-Bit DAC Application.

I. Introduction

A Digital-to-Analog Converter (DAC) is used to convert digital audio signals into an analog signal and DACs are commonly used in music players [1] [2].
There are R-2R ladder DAC, Delta-sigma DAC, and Multibit DAC, and we are designing a Binary Weighted Resistor D/A Converter Circuit, which is based on a summing amplifier. Some use binary, but some don't. [3] [4].

We are designing this DAC with very low cost, as the 3 boards cost only 5 dollars through Oshpark. This is also a great opportunity for beginners to learn how DAC works, and they can design their own PCB to build a simple DAC.

There are five sections in total. The first section talks about the introduction of this report, and the second section shows our SPICE schematic and simulation outputs. The third section presents our PCB designed in Eagle, which is our board layout.

II. 4-BIT DAC Design and Simulation

The first step in the design process after research was designing the circuit schematic, as seen in Figure 1. We chose a 3.01k and a 5.1k Ohm to be combined in series for R1, the highest resistance we needed. For R2, we chose a 4.12k Ohm resistor, a 2.05k Ohm resistor for R3, and a 1k Ohm resistor for R4 and Rf. These values are ones where we can easily find the resistances we need when we build the board. They are also in the correct increments to make as even a drop between steps when the circuit is implemented.

Figure 2 shows the graph of the output voltage of the simulated DAC we will end up building. The even drops between each of the 16 steps show that the DAC will be able to give an accurate analog conversion of any 4-bit signal we give it, since all 16 combinations were tested.

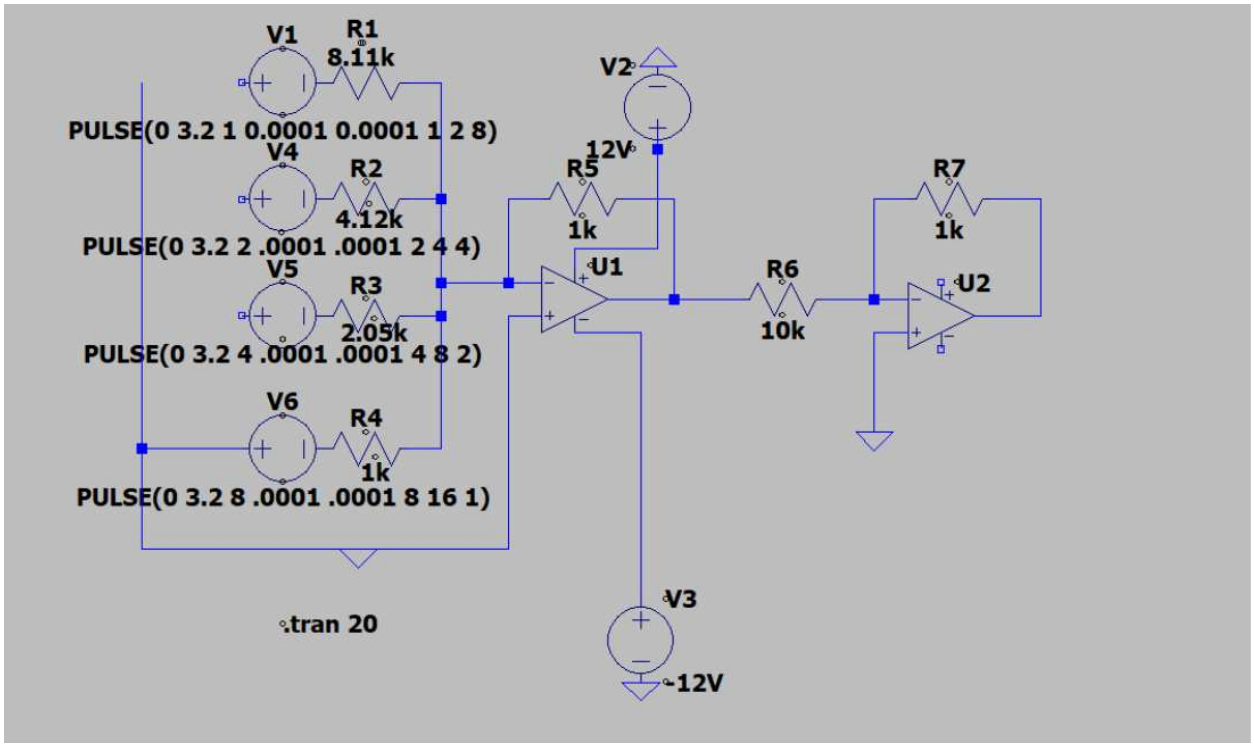


Figure 1: DAC Circuit Schematic

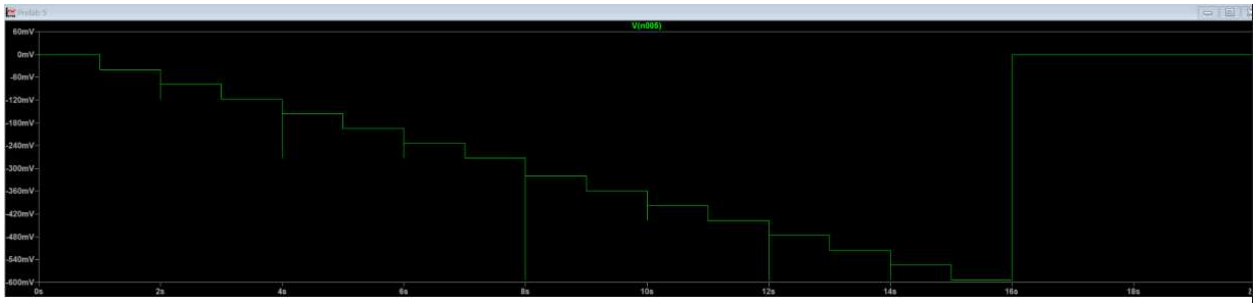


Figure 2: LTSpice DAC Simulation Graph

III. PCB Design

We designed the PCB of the multi-component logic-based circuit in EAGLE version 9.6.2. Subsequently, we prepared the completed and functional board layout for production using the completed schematic. The finished schematic included a number of integrated circuits, passives, test points, and input and output headers. The board was designed within the fabrication limit of “1 by 1”.

In the diagram, all the subcircuits were positioned to enable straightforward routing. The integrated circuits were situated close to associated passive elements in order to minimize trace length. Signal lines were made as brief and linear as practicable in order to preserve signal integrity, and test point locations were provided to enable debugging as well as measurement access for in-lab tests.

In the topography, all the subcircuits were arranged in a way that simple routing was feasible. The ICs were positioned close to associated passive elements for minimizing trace length. The signal paths were made as short and linear as possible for preserving signal integrity, and locations for test points were provided for ease of debugging as well as for access to measurements for in-lab debugging.

We added ground pours to both sides of the board to help reduce electrical noise as well as aid in the dissipation of heat. A series of vias was added to establish good electrical connection between the ground planes. Mounting holes in the four corners of the board have also been added to provide support in the case of future enclosures.

Following the routing process, we ran EAGLE's design rule check (DRC) to double-check against trace width, clearance, and the size of the vias. Once we ensured that the layout was compliant in all respects, the Gerber-formatted layout files were dispatched to OSH Park for production.

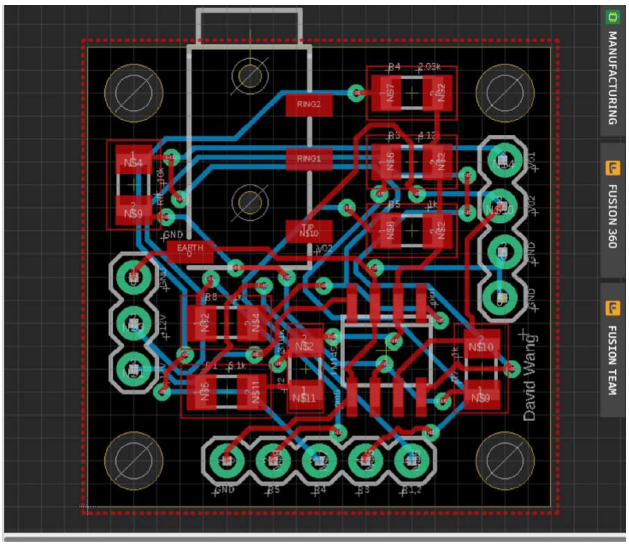


Figure 3: DAC PCB Design

IV. PCB Assembly and Testing

We ordered the 1-inch by 1-inch PCB from OSH Park, which only costs 5 dollars for 3 little boards. However, to build the whole DAC, we also bought the lab kit for it, which costs us 60 dollars. Therefore, the total cost was 65 dollars for the project. To assemble the components onto the PCB, we first used paste and the reflow oven to solder the surface mount resistors and the IC. Then we used the soldering iron to hand-solder the connectors. We used the binary counter to test the PCB, and the output matches the simulations very well.

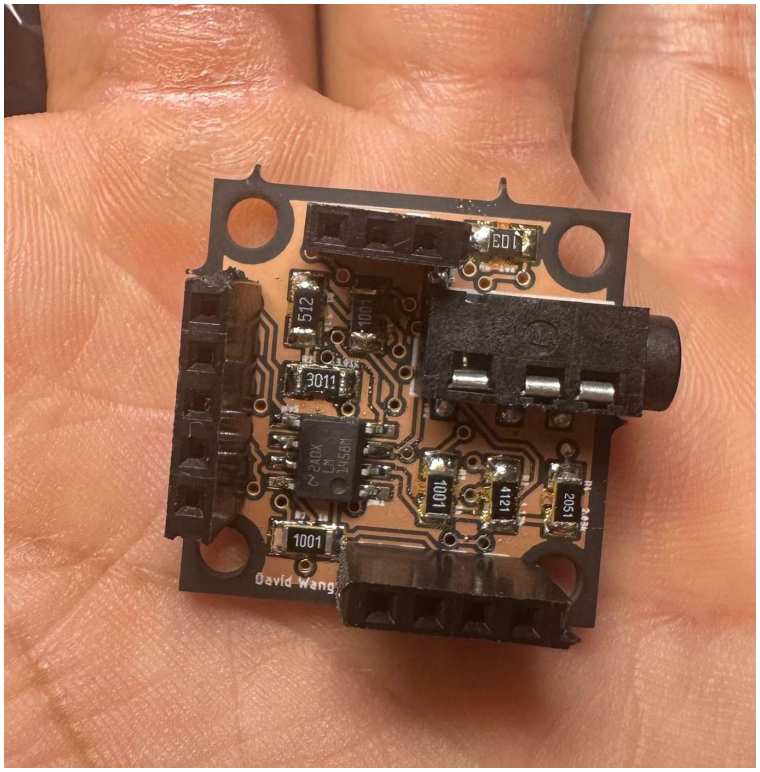


Figure 4: The Assembled PCB

V. 4-Bit DAC Application

We used the DAC circuit we made in series with a level shifter and an Arduino to translate an audio signal from analog to digital and back. The level shifter turns the audio signal into something the Arduino can interpret, and the Arduino turns the analog signal into a 4-bit digital signal. The DAC then transforms this back into an analog signal that can be played through a speaker.

Unfortunately, our implementation did not work. Our level shifter worked as shown by the scope in Figure 5, and our DAC worked as shown by the scope in Figure 6, but the Arduino acting as an analog-to-digital converter did not end up working as shown by the scope in Figure 7. We even tried using multiple Arduinos and tweaking the code. Due to this, we were unable to make a functional electronic device.

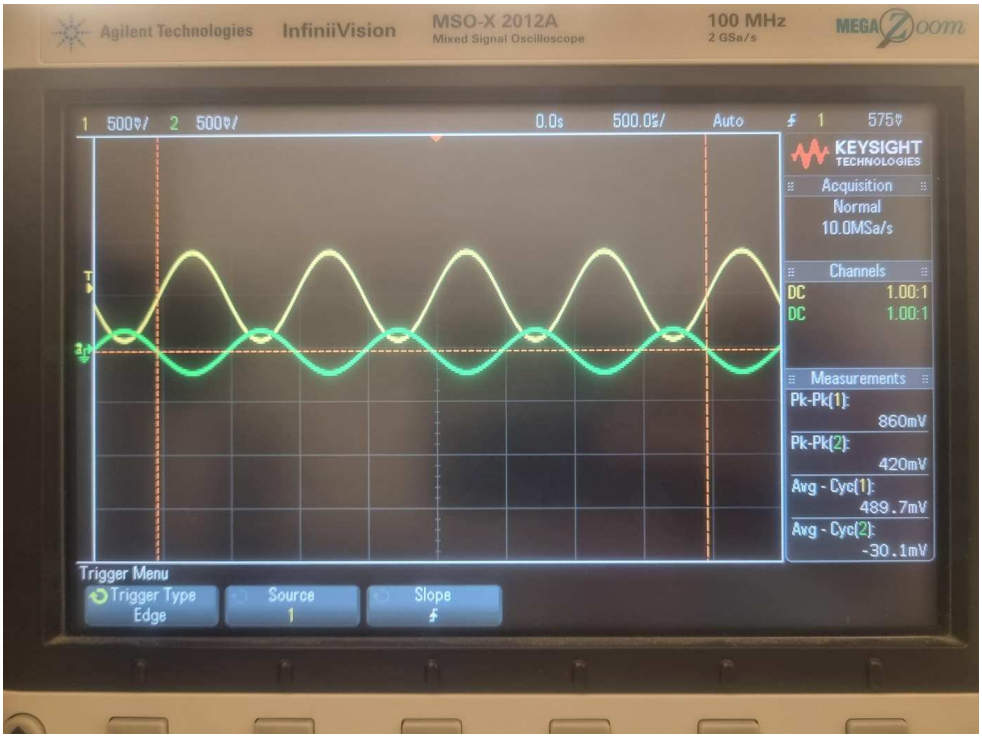


Figure 5: Functional Level Shifter

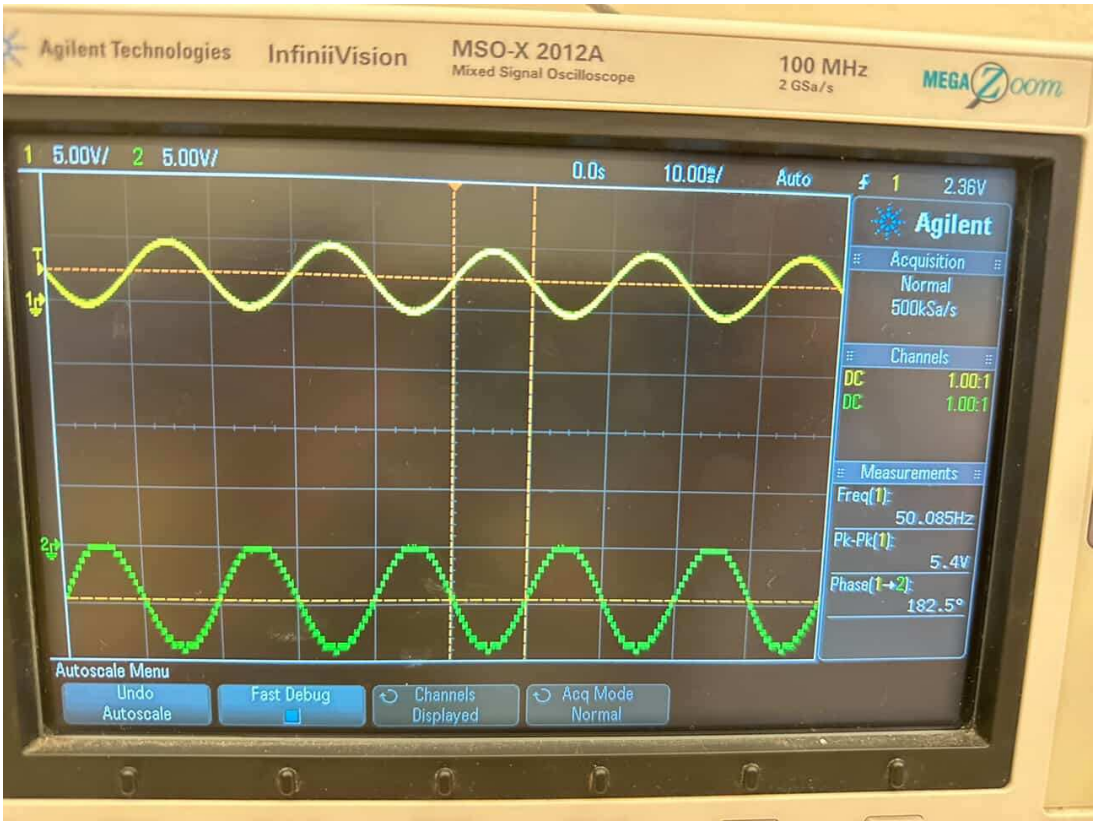


Figure 6: Functional DAC

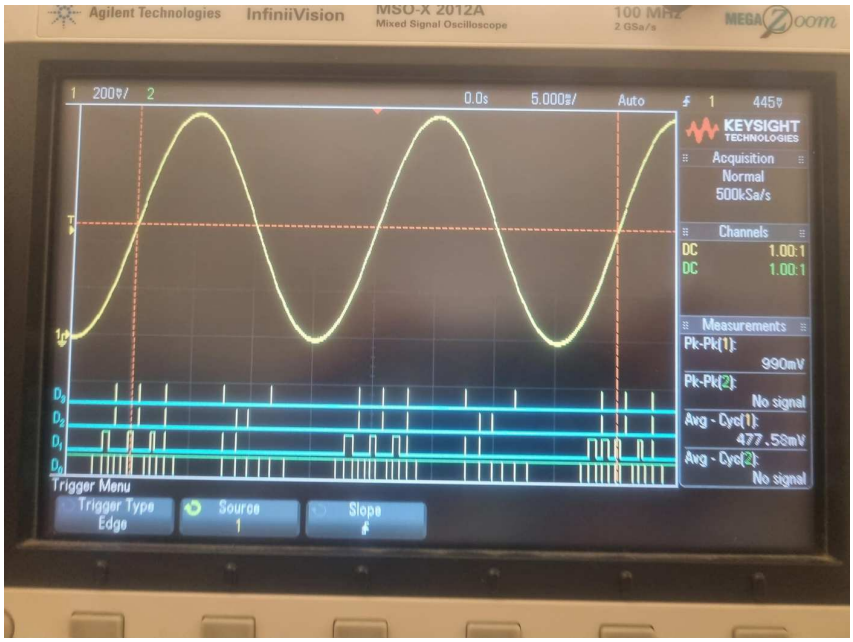


Figure 7: Dysfunctional Arduino

VI. Conclusion

A DAC(Digital-to-Analog Converter) is a type of circuit with a 4-bit input that outputs a digital signal, which is a transformation of that 4-bit analog input. Of the different types of DACs, we designed a weighted resistor D/A Converter Circuit. The design factors we took into consideration the most were cost and simplicity. Our design process and implementation were outlined in the other 5 sections of this report.

References

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